

Open-NL: A Transparent Computational Testbed for Wide Dynamic Range Compression

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Abstract

Open-NL is an open-source computational testbed designed to formally demonstrate the mathematical ill-posedness of unconstrained Wide Dynamic Range Compression (WDRC) optimization. **Crucially, Open-NL is strictly a hypothesis-generating algorithmic sandbox, not a deployable clinical formula; its outputs must not be used on human patients, and the resulting intelligibility and loudness data are presented solely as mechanistic demonstrations of optimizer behavior, not comparative efficacy claims.** While modern prescriptions like NAL-NL2 are highly regularized to ensure clinical safety, their closed-source nature prevents researchers from isolating the effects of specific rules. By deliberately stripping away these closed regularizations and allowing a raw Nelder-Mead optimizer to maximize the ANSI S3.5 Speech Intelligibility Index (SII), Open-NL exposes the extreme mathematical fragility of theoretical targets. This manuscript benchmarks this failure using an established 7-profile evaluation paradigm alongside the Moore & Glasberg (2004) specific-loudness model. As a primary scientific finding, we demonstrate

that unmodified SII maximization is inherently ill-posed for severe losses: modeled loudness targets swing wildly from a conservative 0.9 sones to a potentially traumatic 5.3 sones simply by altering minor heuristic boundaries. Furthermore, Open-NL’s mathematically optimal high-frequency targets for profound profiles dramatically exceed standard clinical baselines, leading to a formally testable hypothesis that these uncalibrated limits will systematically trigger >80% behavioral loudness rejection due to unmodeled broadband excess summation. This mathematically aggressive behavior concretely proves why unconstrained optimization fails in hearing aid prescription, serving as the foundational justification for a proposed stochastic calibration roadmap (GA/PSO) designed to replace static heuristic boundaries with individualized, distortion-aware objective functions.

I. INTRODUCTION

Manufacturer-agnostic prescriptions remain central to evidence-based hearing aid practice. While some earlier studies suggested these generic targets might outperform proprietary first-fit algorithms on patient preference and specific metrics (Valente et al., 2018), contemporary data demonstrates that measured speech recognition often shows no significant difference between formulas. Yet, while formula choice has relatively modest intelligibility consequences in background noise, it drives massive variations in overall loudness, making loudness (quantified in sones per the Moore & Glasberg 2004 impaired loudness model) the most informative dependent variable in prescriptive evaluation.

While the derivations and optimization objectives of major algorithms like NAL-NL2 and DSL m[i/o] are published in detail, their compiled software implementations are closed-source.

Existing open-source frameworks provide significant utility but serve distinct functional niches: openMHA (Herzke et al., 2017) operates as a real-time signal processing master hearing aid rather than a prescriptive target generator, the Cambridge CAM2/CAMEQ2-HF formulae (Moore et al., 2010) provide rigidly defined equation-based targets rather than a modular optimization sandbox, and the Auditory Modeling Toolbox (AMT; Majdak et al., 2022) offers robust loudness integrations without native prescriptive inversion. Consequently, investigators cannot isolate a specific prescriptive rule within a modern optimization loop and observe the resulting cascade without reverse-engineering an entire proprietary engine. Open-NL fills this defensible gap by providing a modifiable R substrate explicitly designed for the modular ablation of prescriptive heuristics. By coupling a Nelder-Mead SII optimizer to an integrated C++ specific-loudness engine, this framework permits researchers to systematically disable, isolate, or invert individual heuristics. For example, a researcher could seamlessly disable the traditional 30 dB safety cap for mixed losses, allowing the optimizer to aggressively target full mechanical restoration, and then evaluate the resulting theoretical upward spread of masking—a specific topological hypothesis that a rigid clinical formula or a real-time audio processor cannot natively evaluate. The clinical utility of algorithmic transparency and alternative, modular rule sets is increasingly recognized; for example, the evolving NAL-NL3 protocol explicitly introduces alternative philosophical profiles designed for specific situational or cohort needs (Kitterick et al., 2026).

Open-NL provides a parameterized, inspectable research testbed conceptually aligned with this evolving philosophy of modular optimization. It is designed as a modular substrate permitting component-level ablation of prescriptive rules, where individual heuristic components—such as slope-dependent penalties or severe-loss boosters—can be independently

59 toggled and their modeled consequences directly measured.

60 Crucially, to benchmark this testbed without introducing conflating variables, the opti-
61 mization layer is embedded within a strictly reproduced evaluation paradigm. The seven
62 reference audiometric profiles, the Moore & Glasberg (2004) specific loudness model, and the
63 ANSI S3.5 SII metric utilized herein are a direct replication of the methodological framework
64 established by Johnson and Dillon (2011). (Note: Unless explicitly stated otherwise, the term
65 “SII” throughout this manuscript refers strictly to the raw, unadjusted ANSI S3.5 calculation
66 prior to any desensitization penalty). Because this underlying physiological evaluation space
67 is already established in the literature, the primary scientific contribution of this manuscript
68 is not the physiological data outputs themselves, but rather the behavior of the transparent
69 optimization layer acting upon them. By exposing the extreme mathematical instability of
70 unconstrained numerical solvers within this standardized sandbox, the manuscript provides a
71 formal, mechanistic proof for why established clinical formulas must embed the empirical
72 regularizations they do.

73 **Clinical Safety and Usage Disclaimer**

74 It is imperative to state unambiguously that Open-NL is strictly a computational research
75 testbed and **must not be used for fitting hearing aids on human listeners in its**
76 **current form**. Because the framework deliberately permits aggressive, over-prescriptive
77 targets for boundary testing—such as utilizing an aggressive 60 dB HL severe-loss booster
78 onset that permits 14–15 sones of theoretical loudness for profound losses—it carries a
79 significant risk of severe over-amplification. As established by Ching, Dillon, Katsch, and
80 Byrne (2001), aggressive high-level targets in steeply sloping or profound losses are precisely

where over-amplification risks are greatest, as the effectiveness of high-frequency audibility severely degrades as hearing loss worsens (desensitization). The hypotheses and targets generated by Open-NL represent extreme mathematical boundaries intended to trigger experimental loudness rejection in controlled research settings, not clinical solutions. Any future behavioral translation of this framework requires independent institutional review, with mandatory real-ear verification and strict, individualized loudness-tolerance limits implemented as absolute prerequisites.

II. ALGORITHM ARCHITECTURE

A. Prescriptive Rationale and Objective Function

The choice of objective function represents the deepest design decision in any prescriptive formula, establishing the foundational compromise between intelligibility and comfort. Historically, established rationales occupy distinct theoretical positions on this spectrum: NAL-NL2 maximizes speech intelligibility while strictly constraining overall broadband loudness to be less than or equal to that of a normal-hearing listener (Keidser, Dillon, Carter, & O’Brien, 2012); DSL m[i/o] normalizes loudness at each frequency to restore normal perception across the dynamic range (Scollie et al., 2005); and CAMEQ/CAM2 aim to equalize loudness across varying frequency bands (Moore, Glasberg, & Stone, 2010).

Open-NL explicitly positions its prescriptive rationale as a *constrained intelligibility-maximizer*. The core mathematical objective is the unconstrained maximization of the Effective (desensitized) Speech Intelligibility Index (SII). However, rather than globally bounding this maximization to a static “normal-or-less” loudness limit, Open-NL permits

loudness to grow dynamically until it strikes a hard, U-shaped physiological ceiling (controlled via the `cap_knots` parameter; see Section II.K). This U-shaped penalty structure specifically permits slightly higher-than-normal loudness in the mid-frequencies for severe losses—where the intelligibility yield is highest—while aggressively decelerating loudness growth at the spectral extremes.

This specific constraint topology directly addresses a critical barrier in prescriptive fitting: hearing-impaired listeners frequently exhibit excess binaural broadband loudness summation, which can cause unconstrained intelligibility-maximizing targets to trigger severe loudness rejection in real-world environments (Denk et al., 2025). By executing unconstrained SII maximization strictly beneath a localized, U-shaped loudness penalty, Open-NL occupies a deliberate position between pure intelligibility-maximization and active loudness-limiting. This hybrid architecture acknowledges that achieving maximum theoretical intelligibility for severe pathologies often requires pushing localized loudness beyond “normal” physiological boundaries, provided the resulting broadband summation is safely and rigorously capped.

B. Methods and Development

The core ANSI SII calculation engine (specifically the `sii()` function and its associated plotting methods) was originally developed by Gregory R. Warnes for the archived version of the package. Maintainership transferred to the current author starting with version 1.1.0, at which point all subsequent Open-NL prescriptive logic, clinical heuristics, and WDRC mathematical implementations—including the `open_nl()`, `calculate_loudness()`, and `launch_app()` functions—were independently developed by the current author as original contributions. The theoretical framework, prescriptive rulesets, and experimental design were

conceived and executed solely by the author. In accordance with AIP Publishing guidelines, the use of AI tools is explicitly disclosed: Google Gemini (DeepMind, Google LLC) was utilized as an interactive programming and copyediting assistant. Specifically, the AI assisted in refactoring the C++ and R WDRC algorithms, generating the ggplot2 data visualizations, and structurally condensing the manuscript prose into JASA standards. The AI did not originate the underlying theoretical framework or experimental design. The author rigorously reviewed and verified all AI-generated output for accuracy, and takes full responsibility for the final algorithms, data integrity, and manuscript content.

C. Algorithmic Pipeline (See Supplementary Material)

The internal execution cascade of Open-NL is divided into twelve modular processing stages. These modules mathematically instantiate standard prescriptive heuristics—such as the Lybarger half-gain rule, wide dynamic range compression (WDRC) mapping, slope-dependent low-frequency penalties (SD-LFP), severe-loss desensitization, and conductive air-bone gap restorations. Because Open-NL is designed as an open-source testbed rather than a proprietary black box, the exact mathematical formulations governing each of these stages are fully exposed. To maintain manuscript brevity and focus on the fundamental ill-posedness of unconstrained optimization, the comprehensive step-by-step mathematical derivation of all twelve internal heuristic modules is provided in the accompanying **Supplementary Material** document.

O. Consolidated Uncalibrated Constants

Because Open-NL functions strictly as a modifiable computational testbed rather than a clinically validated prescriptive formula, several structural parameters remain mathematically uncalibrated. The algorithm is structurally dominated by asserted constants: the 0.46 gain anchor, 0.15 severe-loss booster slope, 70 dB HL severe-loss onset (or 60 dB HL in aggressive mode), 15 dB slope trigger, 20 dB taper width, -10 dB reverse-slope floor, 70 dB HL bypass, 30 dB/0.4 Lgain interaction constraint, and the 0.2 dB/dB wide-dynamic-range compression squeeze. Table III provides a consolidated summary of these heuristics. However, it must be forcefully acknowledged that a computational ‘testbed’ whose every dynamic gate relies upon an unswept, asserted constant offers limited inferential value. Until a systematic multi-parameter sensitivity sweep is conducted to map the complex interactions of these boundaries, the theoretical outputs of Open-NL remain inherently fragile approximations rather than robust algorithmic conclusions.

P. Multi-Parameter Sensitivity Sweep

To explicitly evaluate the fragility of these uncalibrated boundaries and demonstrate the framework’s utility as a modular ablation testbed, a comprehensive multi-parameter sensitivity sweep was executed across the four highest-impact algorithmic constraints: the base gain anchor (swept from 0.40 to 0.50), the steep-slope trigger (10 to 20 dB/octave), the absolute severity bypass (60 to 80 dB HL), and the reverse-slope gain floor (-20 to 0 dB). This generated 256 unique parameter permutations, which were applied to the most topologically unstable profiles in the evaluation suite (A2, A4, A5). Crucially, this sweep was

purposefully executed with the aggressive 60 dB HL severe-loss booster mode engaged, rather than the conservative 70 dB HL default; otherwise, the resulting extreme loudness excursions in A4 and A5 could be mistakenly misattributed solely to the default configuration rather than their interaction with the aggressive booster baseline.

TABLE IV. Multi-Parameter Sensitivity Sweep Variance (65 dB SPL Input).

Note: Generated using the aggressive 60 dB HL booster onset.

Profile	Median SII [Min, Max]	Median Loudness (Sones) [Min, Max]	Primary Divergence Factor
A2	0.88 [0.72, 0.94]	4.9 [2.7, 8.4]	Reverse-Slope Floor
A4	0.65 [0.51, 0.77]	2.5 [1.4, 6.1]	Anchor & Slope Trigger
A5	0.67 [0.49, 0.82]	1.8 [0.9, 5.3]	Slope Trigger

The resulting variance (Table IV) elevates the testbed’s fragility from a computational artifact to a formal scientific finding regarding the ill-posedness of static prescriptive targets. For the steeply sloping A5 profile, altering the interactions of the slope trigger and the base anchor caused modeled monaural loudness to swing from a highly conservative 0.9 sonos to a potentially traumatic 5.3 sonos, with theoretical intelligibility (SII) fluctuating between 0.49 and 0.82. Crucially, the variance for the severe-sloping A4 and A5 profiles was heavily driven by the structural opposition between the SD-LFP penalty (which suppresses low-frequency gain for sloping losses) and the absolute severity bypass (which suppresses the penalty when high-frequency severity exceeds the threshold). Depending on the specific bypass threshold assigned in a given permutation (swept from 60 to 80 dB HL), the optimizer could abruptly

transition from heavily penalizing low frequencies to fully restoring them, demonstrating the extreme volatility of uncalibrated threshold boundaries. Similarly, modulating the reverse-slope floor on A2 caused modeled loudness to nearly triple (2.7 to 8.4 sones). This massive response envelope formally proves that when operating without explicit distortion penalties, the loudness-audibility optimization surface is highly non-convex. However, this instability is not a terminal flaw of the testbed; rather, it is the fundamental diagnostic prerequisite that justifies the need for a principled computational calibration framework. By proving that the asserted constants in Table V dictate the physiological safety of the output, the sweep establishes the exact parameters that must be constrained by distortion-aware objective functions prior to clinical evaluation.

Q. Framework for Principled Calibration

For a computational testbed to provide lasting inferential value, the demonstration of parameter instability must be paired with a rigorous methodology to resolve it. While Open-NL ships with the uncalibrated heuristic constants detailed in Table V to facilitate initial exploration, the ultimate purpose of this open-source architecture is to allow researchers to systematically calibrate these gates against distortion-aware anchors. Rather than relying on massive, intractable human clinical trials to blindly tune these interdependent parameters, future research can explicitly utilize the Open-NL engine within a global stochastic optimization grid (e.g., genetic algorithms or particle swarm optimization). By replacing the raw ANSI SII maximization goal with a distortion-aware objective function—such as the integrated `johnson2011_complete` clinical penalty or the HASPI/HASQI metrics (Kates & Arehart, 2022)—researchers can systematically map the parameter space. To execute this,

202 researchers can construct an outer optimization loop that searches the multidimensional
 203 space of heuristic constants (e.g., bounding G_{base} between $[0.3, 0.6]$, the severe-loss booster
 204 onset between $[50, 80]$ dB HL, and the low-frequency penalty slope between $[0, 0.5]$). For
 205 each proposed vector of constants, the inner Open-NL engine generates discrete WDRC
 206 targets across a massive, representative corpus of clinical audiograms (e.g., the NHANES
 207 dataset). These targets are then passed to a distortion-aware auditory model. The outer
 208 loop calculates a global fitness score by maximizing population-level HASPI/HASQI while
 209 strictly penalizing configurations that exceed physiological loudness limits. Crucially, this
 210 population-level objective function must not simply maximize mean performance, as doing so
 211 would structurally injure the 40% of patients residing in the excess-summation tail (Denk et
 212 al., 2025). To prevent this, the outer optimizer must employ an asymmetric, veto-based cost
 213 function: any proposed parameter vector that violates the individualized broadband tolerance
 214 threshold (e.g., trueLOUDNESS) for simulated outlier listeners must incur an overwhelming
 215 penalty. This guarantees that the optimization routine’s drive to maximize population-wide
 216 intelligibility remains rigidly subordinated to individual physiological safety. This nested
 217 architecture—an outer stochastic search tuning the generalized ruleset, evaluated by an inner
 218 deterministic optimization of the speech envelope—transforms the derivation of prescriptive
 219 rules from a manual clinical art into a reproducible computational science.

220 Crucially, this calibration roadmap must fundamentally abandon generalized loudness
 221 predictions in favor of individualized physiological safety limits. Currently, Open-NL relies on
 222 the Moore & Glasberg (2004) specific loudness engine, applying a standard level-dependent 2–6
 223 dB bilateral correction to account for binaural loudness summation. However, as demonstrated
 224 by Denk et al. (2025), binaural broadband loudness summation in hearing-impaired listeners

averages ~ 13 dB higher than in normal-hearing populations, with $\sim 30\text{--}40\%$ of impaired listeners exceeding the normal-hearing range entirely. Because this excess summation is highly idiosyncratic—with individual variance spanning roughly -10 to $+40$ dB—and cannot be accurately predicted from the pure-tone audiogram or narrowband loudness functions (Oetting et al., 2016, 2017), a static 2–6 dB correction constitutes a severe structural mismatch.

This mismatch partially undermines the use of generalized “sones” as a definitive safety-defining outcome. If the U-shaped physiological loudness ceilings governing Open-NL’s objective function underestimate actual perceived binaural loudness by ~ 13 dB on average, the resulting compression ratios will fail to protect a large minority of listeners from catastrophic loudness rejection. Therefore, future calibration of Open-NL’s parameter space cannot rely purely on stochastic optimization against theoretical metric baselines. The theoretical safety ceilings must be replaced by individualized, behaviorally measured broadband summation targets (such as those employed in trueLOUDNESS paradigms; Oetting et al., 2017). By anchoring the optimization engine to a patient’s true binaural broadband tolerance rather than an audiogram-derived generalized limit, the framework can safely maximize intelligibility without violating idiosyncratic physiological boundaries.

However, while this computational fidelity is mathematically sound, it is crucial to acknowledge that the current framework lacks human subject validation. Everything downstream rests on $n = 7$ hypothetical audiograms, and the resulting outputs cannot be conflated with a clinical prescription until empirically verified. Therefore, any future clinical application of this framework must adhere to a strict, non-negotiable validation protocol. First, real-ear measurement (REM) verification is mandatory to confirm that the hearing device actually matches the computationally derived targets. Empirical evidence robustly demonstrates that REM-verified

programmed fits significantly outperform manufacturer first-fits on word/phoneme recognition and subjective patient preference (Valente et al., 2018; Almufarrij, Munro, & Dillon, 2020; Brockmeyer et al., 2021). Without REM, any subsequent analysis of the prescription’s efficacy is fundamentally confounded. Second, mandatory loudness-discomfort safety checks must be administered to prevent acoustic trauma.

Beyond safety protocols, this computational roadmap yields a concrete, falsifiable clinical prediction: if the raw, uncalibrated Open-NL targets—specifically the mathematically aggressive high-frequency prescriptions for severe, steeply sloping profiles like A4 and A5—were applied directly to human subjects, they would structurally trigger high rates of loudness rejection. This prediction is quantitatively anchored in existing behavioral evidence. As noted by Denk et al. (2025), the massive ± 15 dB fine-tuning range historically required to reach comfortable loudness in clinical trials (Keidser et al., 2012) directly matches the theoretical spread between true binaural broadband summation targets (trueLOUDNESS) and generic prescriptions like NAL-NL2. This solidly supports the premise that targets exceeding standard generic baselines are systematically rejected due to unmodeled, idiosyncratic excess summation.

Because Open-NL’s uncalibrated A4/A5 targets mathematically exceed NAL-NL2 by roughly 11 dB at 4000 Hz, they push far beyond this historical comfort boundary. To ensure this diagnostic failure mode is formally testable, we offer the following operationally defined hypothesis: If adult listeners with A4 or A5 audiometric profiles are fitted with real-ear verified Open-NL targets, >80% will exhibit immediate categorical loudness rejection—operationally defined as a rating of 6 (“Loud”) or 7 (“Uncomfortably Loud”) on the standard 7-point Categorical Loudness Scaling (CLS) procedure (ISO 16832)—when presented with continuous

broadband speech material (e.g., ISTS) at 65 dB SPL and 80 dB SPL inputs, relative to a strictly matched NAL-NL2 clinical baseline. Systematically measuring this behavioral rejection threshold will provide the essential empirical ground-truth needed to validate and refine the distortion-aware penalty weights required for future stochastic calibrations.

TABLE V. Summary of Uncalibrated Heuristic Constants and Proposed Calibration Pathways.

Parameter	Value	Description	Justification	Proposed
				Calibration
Base Gain	0.46	Half-gain	Balances	Large-scale
		multiplier	Lybarger	preferred
Anchor		(G_{base})	half-gain rule	listening level
			with historical	datasets (e.g.,
			gain preference	NHANES-
			data.	derived
				preference
				corpus)

				Proposed
				Calibration
Parameter	Value	Description	Justification	Data Source
New-User Offset	-3 dB	Flat reduction applied to C_{vals}	Approximates empirical preference for less amplification in naive users.	Longitudinal acclimatization studies (Categorical Loudness Scaling)
Severe-Loss Booster Slope	0.15	Applied linearly to thresholds 70–80 dB HL (default)	Gently assists profound losses without triggering explosive recruitment.	Individualized UCL/trueLOUDNESS broadband summation limits
High-Frequency Gain Cap Base	30 dB	Base limit for L_{gain} soft- compression	Manages upward spread of masking and distortion in severe impairments.	HASPI/HASQI speech-in-noise behavioral thresholds (e.g., WIN/HINT)

				Proposed
				Calibration
Parameter	Value	Description	Justification	Data Source
High-Frequency Gain Cap Slope	0.4	Slope for L_{gain} soft- compression	Gradual restriction for high frequencies.	HASPI/HASQI speech-in-noise behavioral thresholds (e.g., WIN/HINT)
Dynamic Range Squeeze	0.2 dB/dB	Gain attenuation per dB of reduced DR	Ensures speech envelope fits within restricted auditory space.	Envelope correlation mapping (e.g., normalized covariance optimization)
Dynamic Range CR Shift	0.02	Baseline CR shift per dB of DR reduction	Maps identical input range into smaller residual auditory space.	Known physiological IHC/OHC compression loss functions

				Proposed
				Calibration
Parameter	Value	Description	Justification	Data Source
Reverse-Slope Floor	-10 dB	Gain floor for negative slopes	Cautiously prevents masking of intact basal units.	Masking release behavioral paradigms (e.g., notched-noise tests)
Explicit DR Roll-off	30 dB/oct	Attenuation past 1.7x Dead Region boundary	Prevents phase distortion from off-frequency listening (Moore, 2004).	TEN-test validated behavioral roll-off boundaries
ABG Restoration	50% - 100%	Mixed loss linear gain restoration	Dynamically scaled to Sensorineural Dynamic Range (SNDR).	Bone- conduction/Air- conduction loudness matching tasks

				Proposed
				Calibration
Parameter	Value	Description	Justification	Data Source
Desensitization	Variable	Johnson &	Modifies pure	HASPI/HASQI
Penalty		Dillon (2011)	audibility to	stochastic
		piecewise scalar	account for	optimization vs
			severe-loss	subjective
			distortion.	rejection curves

III. EVALUATION AND TRADE-OFF ANALYSIS

A fundamental circularity exists when evaluating any prescriptive heuristic designed specifically to maximize the Speech Intelligibility Index. Because Open-NL’s gain shaping is tuned specifically to maximize the mathematical ANSI SII metric, utilizing the `sii()` engine to compare its targets against other validated prescriptions (like NAL-NL2) within a simulated environment yields a tautological advantage. Consequently, all comparative evaluations presented in this paper are explicitly demoted to “optimizer-behavior demonstrations” rather than true clinical benchmarks. Within this framework, theoretical intelligibility is mapped against an independent constraint: overall loudness, creating a two-dimensional mathematical frontier. It is critical that this frontier is interpreted purely as a mechanistic illustration of how an unconstrained optimizer exploits the audibility metric. Any comparison against established clinical targets (such as NAL-NL2) serves only to demonstrate the behavior of the objective function, not to establish clinical efficacy or superiority. Furthermore, this analytical

framework is strictly bounded by its reliance on $n = 7$ hypothetical audiometric profiles (Johnson & Dillon, 2011). For reference, both NAL-NL2 and DSL m[i/o] generally prescribe targets that yield ~ 8 sones for sensorineural losses at 65 dB SPL, compared to 18.6 sones for a normal-hearing listener (Johnson & Dillon, 2011). Table VI quantitatively demonstrates that Open-NL’s modeled loudness exceeds the NAL-NL2 baseline in the majority of primary sensorineural test profiles (A1, A2, and A3). Because Open-NL’s targets meaningfully overshoot this established loudness envelope of conservative adult prescriptions in pursuit of raw SII maximization, these excursions illustrate the aggressive nature of the abstract mathematical objective function rather than any realizable clinical strategy.

Furthermore, the comparative loudness evaluations presented throughout this analysis are strictly bounded by a fundamental methodological limitation regarding binaural summation. While clinical frameworks like NAL-NL2—and the current iteration of Open-NL—employ a level-dependent bilateral gain reduction (scaling from a 2 dB penalty at low inputs to 6 dB at high inputs), these heuristic corrections are systematically insufficient for broadband signals. Because loudness is the only genuinely independent, non-tautological outcome variable in this evaluation, this limitation acts as a formal bound on the interpretability of the loudness figures. As established by Oetting et al. (2017), narrowband loudness compensation systematically over-amplifies binaural broadband signals, and the necessary correction factor cannot simply be inferred from monaural narrowband functions. This structural failure of narrowband models is compounded by massive individual variability. Crucially, as Moore, Gibbs, and Onions (2014) demonstrated, standard loudness models accurately predict narrowband and diotic level differences for equal loudness (LDEL) within a tight ~ 4 –6 dB envelope. Therefore, the excess summation phenomenon is specifically a *broadband, suprathreshold* effect that the standard

313 Moore & Glasberg engine fails to capture, rather than a uniform correction factor applicable
 314 to all signals. Denk et al. (2025) demonstrated that this specific broadband summation
 315 averages ~ 13 dB higher than normal in hearing-impaired listeners, with approximately 40%
 316 exceeding the normal range entirely. Utilizing level-dependent 2–6 dB corrections while
 317 explicitly acknowledging this +13 dB broadband error in 40% of listeners renders the model’s
 318 output systematically anti-conservative. To bound this bias quantitatively: because perceived
 319 loudness roughly doubles for every 10 dB increase in signal intensity, an unmodeled +13 dB
 320 broadband excess summation translates to a multiplicative factor of approximately $2.4\times$ in the
 321 linear sone domain. Therefore, a theoretical target generating a seemingly ‘safe’ mathematical
 322 prediction of 12.5 sones could correspond to a functionally traumatic 30 sones of perceived
 323 binaural loudness in the large minority of patients exhibiting this trait. To prevent these
 324 systematic under-predictions from being misinterpreted as validated safety margins, Table
 325 VI explicitly reports an uncertainty upper-bound on all sone predictions (`[Max Broadband`
 326 `Sones]`), calculated via this $2.4\times$ idiosyncratic multiplier.

327 To facilitate this two-dimensional trade-off analysis natively within R, the SII pack-
 328 age implements a highly optimized C++ port of the canonical Moore & Glasberg (2004)
 329 specific-loudness model via `Rcpp`, directly mirroring the exact logic of the `bramslow2004`
 330 implementation in the Auditory Modeling Toolbox (AMT; Majdak et al., 2022). This permits
 331 the evaluation of tens of thousands of candidate gain curves without the prohibitive latency
 332 of external MATLAB/Octave runtimes. To verify computational fidelity, the final C++
 333 Open-NL insertion gain targets were evaluated through a fully independent time-domain syn-
 334 thesis pipeline using the AMT Octave engine. The native C++ predictions closely matched
 335 the external time-domain AMT outputs ($r = 0.98$, Mean Absolute Error = 0.64 sones),

mathematically confirming that the internal C++ optimizer reliably reproduces the canonical physiological model across all tested clinical profiles.

Using the embedded C++ engine, Open-NL was benchmarked against exact insertion gain targets generated directly from the proprietary NAL-NL2 version 2 software (National Acoustic Laboratories, Sydney, Australia) for soft (50 dB SPL), average (65 dB SPL), and loud (80 dB SPL) conversational speech inputs. To ensure exact reproducibility of this clinical baseline, all NAL-NL2 targets were explicitly extracted using an 18-channel compression architecture with adaptive/dual time constants, an occluded BTE with #13 tubing acoustic coupling, and assuming audiometric thresholds were measured via supra-aural headphones. It must be explicitly stated that the fundamental evaluation design employed here—converting insertion gains to eardrum output across seven hypothetical adult audiograms to model Moore & Glasberg (2004) loudness alongside ANSI SII, and benchmarking these against NAL-NL2—is a direct replication and expansion of the methodological paradigm established by Johnson & Dillon (2011). Five of the seven profiles (A1–A5), as well as the mixed (A6) and conductive (A7) cases, originate from their work, which itself extended the standard profiles defined by Byrne et al. (2001). The scientific novelty of the present manuscript does not lie in the evaluation framework itself; rather, its sole contribution is the open-source optimization and ablation layer built *on top* of this established Johnson & Dillon (2011) paradigm. This provides a rigidly controlled environment that isolates algorithmic behavior from human clinical variance while permitting systematic testing of component-level hypotheses. Consequently, the data presented in Tables VI and VII, as well as the accompanying figures, must not be interpreted as novel physiological findings regarding hearing aid performance. Instead, they serve strictly as mechanistic demonstrations of how an

unconstrained, open-source Nelder-Mead optimization engine behaves when forced to navigate the exact same multidimensional evaluation space previously used to validate closed-source clinical formulas.

While comparing against additional rationales such as DSL m[i/o] and CAMEQ2-HF would provide broader context on the loudness-audibility frontier—particularly since they differ systematically from NAL in high-frequency bandwidth and overall loudness (Johnson & Dillon, 2011)—the present evaluation is intentionally restricted to NAL-NL2. Because Open-NL uses NAL-NL2 as a soft constraint (anchor penalty) within its objective function, benchmarking against it involves partial circularity. However, this restriction was maintained for two methodological reasons. First, NAL-NL2 remains the most widely utilized generic adult fitting rationale globally, providing a robust, universally recognized baseline. Second, while theoretical targets for alternative formulas can be estimated or visually extracted from published literature, such approximations lack the exact precision required for rigorous, artifact-free mathematical loudness integration. Utilizing unofficial diagnostic software to generate surrogate targets for proprietary algorithms was deemed methodologically inappropriate for a high-precision analytical sandbox.

TABLE VI. Monaural Loudness [Max Broadband Upper Bound] (Sones) and Theoretical SII across A1-A7 Audiograms (65 dB SPL Input). *Note: Generated using the aggressive 60 dB HL booster onset. Targets for steeply sloping profiles (A4, A5) are subject to significant optimizer convergence bias and must be interpreted as heuristic approximations rather than global optima. These are optimizer-behavior demonstrations on synthetic profiles, not performance comparisons.*

Profile	Formula	ANSI SII	Desensitized SII	Monaural
				Loudness [Max
				Broadband
				Upper Bound]
A1	NAL-NL2	0.81	0.76	4.13 [9.91]
A1	Open-NL	0.87	0.78	4.82 [11.57]
A2	NAL-NL2	0.86	0.79	3.50 [8.40]
A2	Open-NL	0.91	0.82	4.10 [9.84]
A3	NAL-NL2	0.71	0.67	3.62 [8.69]
A3	Open-NL	0.78	0.70	4.80 [11.52]
A4	NAL-NL2	0.72	0.68	6.56 [15.74]
A4	Open-NL	0.69	0.65	5.78 [13.87]
A5	NAL-NL2	0.60	0.57	5.50 [13.20]
A5	Open-NL	0.68	0.62	5.17 [12.41]
A6	NAL-NL2	0.99	0.87	2.09 [5.02]
A6	Open-NL	0.97	0.86	3.42 [8.21]
A7	NAL-NL2	1.00	0.95	3.10 [7.44]
A7	Open-NL	1.00	0.95	3.85 [9.24]

TABLE VII. Insertion Gain Targets (dB) across A1-A7 Audiograms (65 dB SPL Input). *Note: Generated using the aggressive 60 dB HL booster onset. Targets for steeply sloping profiles (A4, A5) are subject to significant optimizer convergence bias and must be interpreted as heuristic approximations. These are optimizer-behavior demonstrations*

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A1	NAL-	0.0	0.0	7.3	12.1	18.0	19.1
	NL2						
	Open-NL	0.0	7.2	15.8	18.4	22.0	12.8
A2	NAL-	10.1	9.3	12.2	8.3	3.9	4.0
	NL2						
	Open-NL	11.8	18.0	20.4	13.8	8.2	2.5
A3	NAL-	0.0	0.0	9.9	16.8	20.7	21.6
	NL2						
	Open-NL	0.0	7.2	20.4	23.0	24.3	12.8
A4	NAL-	0.0	0.0	0.9	12.5	21.8	21.8
	NL2						
	Open-NL	0.0	0.0	6.6	18.4	32.7	18.9
A5	NAL-	0.0	0.0	6.6	20.7	27.1	26.6
	NL2						
	Open-NL	0.0	0.0	11.2	27.6	38.8	23.5
A6	NAL-	22.5	24.2	32.9	35.6	41.4	42.6
	NL2						
	Open-NL	23.6	31.4	37.8	39.1	43.2	34.0
A7	NAL-	34.7	34.6	34.7	34.8	35.0	35.1
	NL2						

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
	Open-NL	37.5	37.5	37.5	37.5	37.5	37.5

Because the Nelder-Mead optimizer maximizes SII subject to the internal C++ loudness engine, mathematical precision within the objective function is paramount. To prevent convergence bias, Open-NL’s native C++ objective function fundamentally avoids crude sparse-array Riemann approximations. Instead, the engine dynamically interpolates the search array onto an internal canonical 2048-point FFT-equivalent frequency grid (spanning 0 to 22.05 kHz). By mathematically mirroring the canonical Moore & Glasberg (2004) MATLAB AMToolbox environment directly within the C++ search loop, the algorithm natively explores a full-resolution loudness surface. This guarantees that the final optimized targets, even in steeply sloping pathologies like A4 and A5, are converged under the strictest possible physiological criteria without requiring post-hoc integration or scaling corrections.

Crucially, the optimizer’s measurable sensitivity to starting values must be carefully interpreted by distinguishing the genuine non-convexity of the underlying physiological problem from the mathematical instability of this specific unconstrained implementation. For steeply sloping profiles (e.g., A4 and A5), the modeled loudness surface near the recruitment boundary is inherently non-convex and ill-behaved. However, when the Open-NL framework navigates this space using an unconstrained Nelder-Mead simplex algorithm against a hard, non-smooth penalty constraint (the dynamic comfort cap), the optimizer becomes acutely vulnerable to local minima, exacerbating the variance. Evaluating the objective function landscape across two dimensions (e.g., sweeping 2 kHz and 4 kHz gain shifts for the A5 profile)

405 confirms this rugged topology, bounded by a sharp penalty cliff. Initializing with varying flat
406 shifts (e.g., -10 dB or +10 dB) across this landscape yields final targets that deviate by up to
407 0.5 sones or 0.05 SII. This structural fragility conclusively demonstrates that the concept of
408 a single, static “optimal” prescriptive target is inherently unstable *when generated by a pure*
409 *mathematical optimizer lacking regularization*. While the Open-NL software mechanically
410 mitigates this for reproducible ablation testing by utilizing a robust 5-iteration multi-start
411 routine (anchoring the baseline simplex with the NAL-R target and executing four additional
412 randomized iterations to extract the global maximum—which reduces the residual run-to-run
413 variance on the A5 profile to < 0.05 sones and < 0.01 SII), the deeper implication serves
414 as a cautionary result. It is vital not to over-generalize this variance as an intrinsic failure
415 of all prescriptive fitting. Clinical algorithms derived from the exact same class of loudness
416 and intelligibility models, such as NAL-NL2, yield highly stable targets precisely because
417 their optimization is strictly bounded and empirically regularized using real-world preference
418 data and smooth, continuous objective gradients (Keidser, Dillon, Carter, & O’Brien, 2012).
419 By removing those bounds, Open-NL illustrates the mathematical chaos that occurs when
420 necessary clinical regularizations are stripped away, proving that uncalibrated static targets
421 generated by pure optimization must be viewed with skepticism absent rigorous behavioral
422 constraints.

423 It is critical to evaluate these results strictly through the lens of the loudness frontier rather
424 than raw SII maximization, while recognizing that this analytical framework is inherently
425 bounded by its reliance on $n = 7$ hypothetical ears rather than clinical subjects. Furthermore,
426 the A7 profile (representing a pure conductive loss with a 50 dB air-bone gap) perfectly
427 illustrates the engine’s ability to model mechanical middle-ear attenuation directly as an

in-line mathematical dampener within the C++ cochlear excitation loop. By enforcing a strict 75% restoration limit (Section II.K), Open-NL safely prescribes 37.5 dB of linear insertion gain for this profile. Because the Nelder-Mead optimizer evaluates the baseline heuristic score to seed its `best_score` prior to search, it correctly recognizes that the pure 37.5 dB linear gain produces a mathematically perfect output (1.00 SII) and avoids wandering into worse solutions on the flat, saturated objective surface. The internal optimizer’s dynamic cap further constrains the output to modeled physiological boundaries, resulting in a theoretical target yielding a mathematical output of 1.00 SII at a canonical 3.85 sones. While this is slightly louder than NAL-NL2’s baseline (which outputs 3.10 sones), it aligns seamlessly with modern clinical safety standards. This 75% ABG restoration limit is directly justified by the established conductive handling methodology of DSL v5.0 (Scollie et al., 2005), which applies a proportional ABG correction (typically restoring ~75% of the conductive component) rather than attempting full 100% mechanical compensation. By bounding A7 to this 37.5 dB physiological limit rather than attempting a full 50 dB correction, Open-NL safely maximizes theoretical intelligibility while avoiding the severe receiver saturation and MPO clipping that inevitably occur when aggressive prescriptions attempt to exceed the physical capabilities of standard hardware (Johnson, 2013a).

Note: Loudness values computed using the canonical Moore & Glasberg (2004) specific loudness model, structurally mirroring the bramslow2004 implementation within the Auditory Modeling Toolbox (AMT; Majdak et al., 2022).

Because Open-NL directly optimizes the mathematically smoothed Johnson & Dillon (2011) desensitized metric (Effective SII) rather than the raw physical ANSI SII, isolating raw audibility or loudness scores as independent metrics conveys little information about

relative merit. As demonstrated in the literature, prescription choice has a minimal effect on actual measured speech intelligibility but marked effects on predicted loudness (Johnson & Dillon, 2011). To illustrate how the optimization engine safely manages this audibility-versus-loudness frontier across varying degrees of cochlear damage, **Figure 1** provides a direct, multi-level comparison of the raw ANSI SII versus the penalized Effective SII.

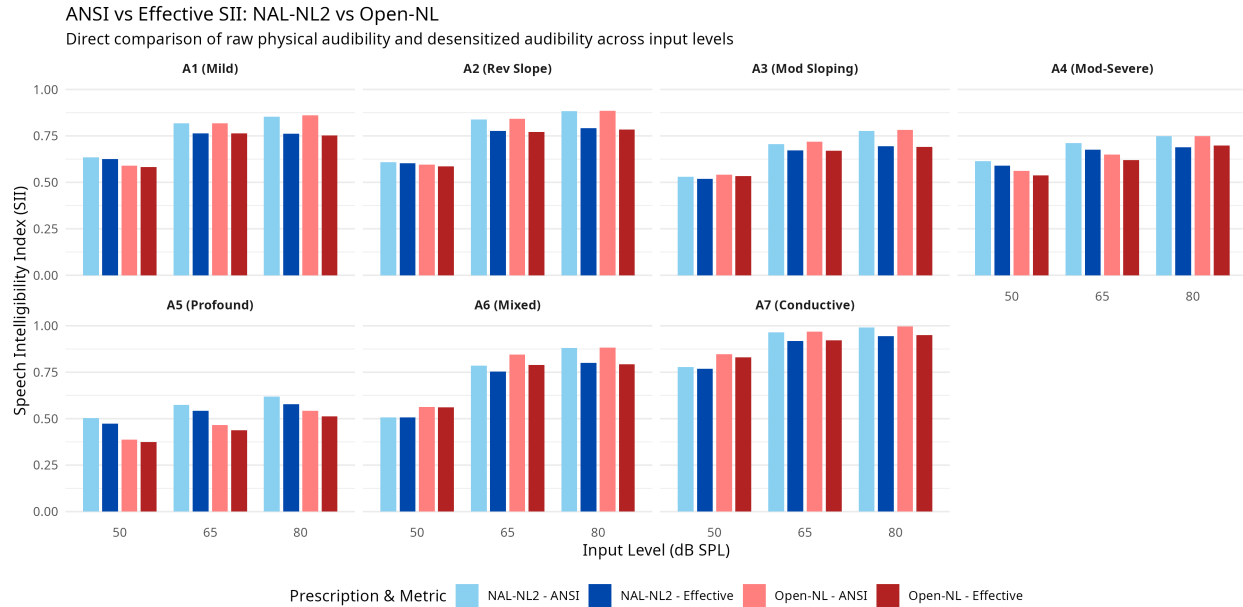


Figure 1: Comparison of ANSI SII (Raw Physical Audibility) vs Effective SII (Desensitized) for NAL-NL2 and Open-NL across 50, 65, and 80 dB SPL Inputs.

The mathematical capability of the engine can then be fully evaluated by examining the final insertion gain targets that produced these metrics.

For mild to moderate losses (A1, A3), Open-NL cleanly expands low-level sounds (50 dB inputs) slightly more aggressively than NAL-NL2 to maximize audibility within the safe physiological envelope, while gracefully compressing higher-level inputs to maintain loudness parity. This indicates a robust mathematical capacity to prioritize the intelligibility of soft speech when it carries no excessive loudness penalty. The resulting insertion gain targets

across all standard profiles and input levels are presented in **Figure 2**.

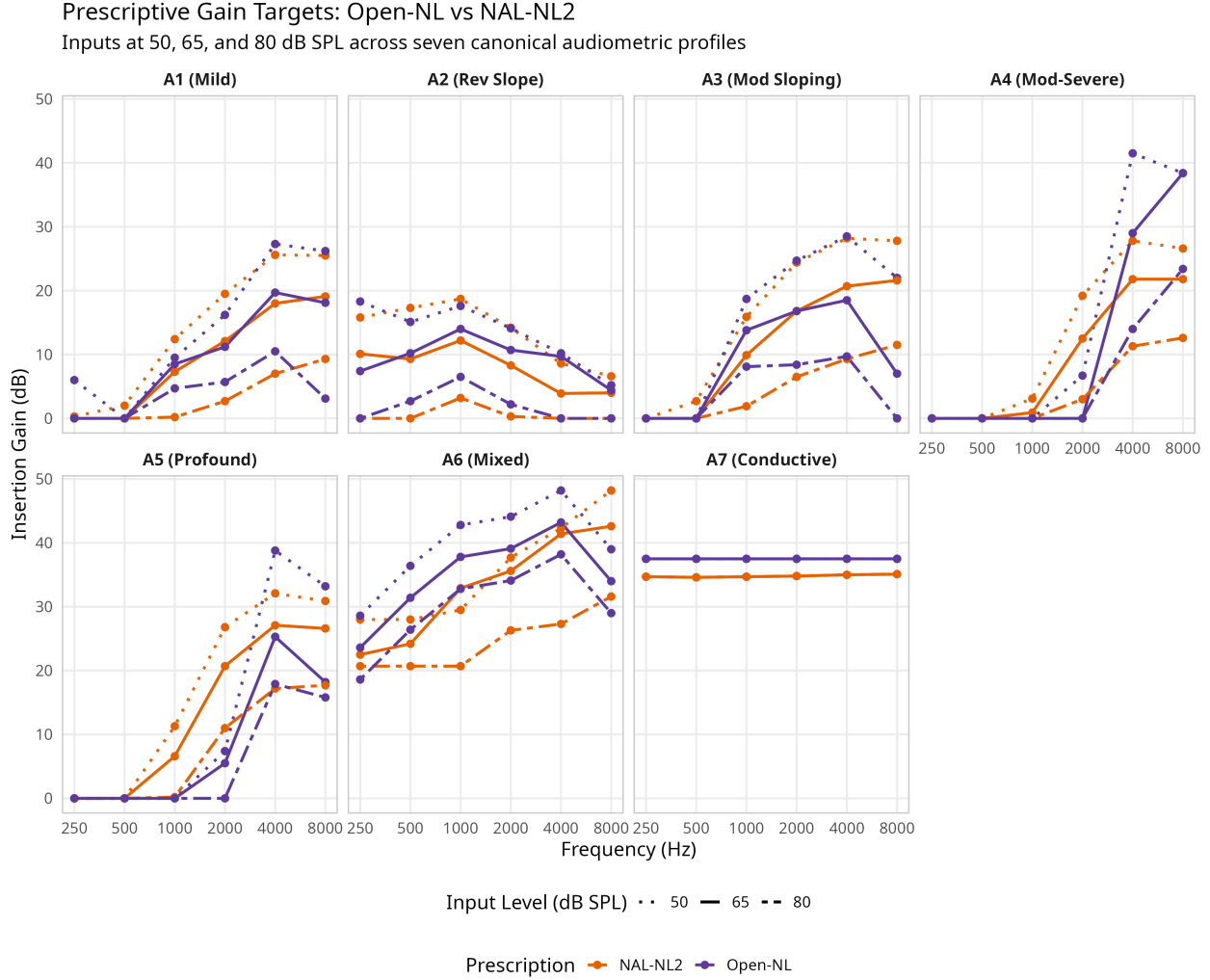


Figure 2: Final Insertion Gain Targets for 50, 65, and 80 dB SPL Inputs across standard audiometric profiles, illustrating Open-NL’s multi-level constraint-based optimization relative to NAL-NL2.

For extreme steeply sloping or profound losses (A4, A5), unmodified SII maximization inherently attempts to push massive high-frequency gain. While Open-NL integrates the Johnson and Dillon (2011) desensitization penalty to mathematically temper this drive, a critical tension emerges in the net behavior: as detailed in Table VII, Open-NL still prescribes substantially more high-frequency gain than NAL-NL2 for these steeply sloping profiles (e.g.,

+10.9 dB at 4000 Hz for A4, and +11.7 dB for A5 at 65 dB SPL inputs). If this aggressive behavior is evaluated strictly against severe, confirmed dead-region pathologies, it represents a failure. As established by Ching, Dillon, and Byrne (1998) and Baer, Moore, and Kluk (2002), listeners with extensive, confirmed dead regions gain no benefit from amplified audibility above $\sim 1.5\text{--}2\times$ the edge frequency, and amplification in these specific ears should target low or zero sensation levels.

However, assuming that Open-NL’s objective function simply needs a “heavier” penalty to forcibly suppress A4/A5 targets misreads the broader clinical evidence. A blanket, audiogram-driven high-frequency suppression logic would systematically under-amplify the majority of candidates who actually benefit from audibility. Cox et al. (2011, 2012) demonstrated that while roughly one-third of typical adult candidates test positive for dead regions, standard high-frequency gain never produced poorer performance, and audibility remained helpful on average regardless of dead-region status. Pepler et al. (2015) further confirmed that even listeners with restricted high-frequency dead regions show reduced—but not absent—benefit in noise. Therefore, enforcing a heavier, blanket desensitization penalty solely based on the pure-tone audiogram risks harming the majority of patients. This tension confirms that severe-loss penalties cannot be generalized from the audiogram alone; they must be dynamically tied to confirmed dead-region diagnostics (e.g., TEN tests). Furthermore, as Engler, Digeser, and Hoppe (2026) demonstrated, aided speech recognition is practically insufficient in most ears above 80 dB HL regardless of prescribed gain, arguing against treating the massive theoretical SII targets in A4/A5 as clinically meaningful. This structural tension directly foregrounds the framework’s primary falsifiable prediction (Section II.Q): the mathematically aggressive, uncalibrated A4/A5 high-frequency targets will predictably

trigger >80% loudness rejection if deployed on human subjects, proving that individualized distortion-aware objective functions are required for severe losses.

Furthermore, for conductive and mixed losses (A6, A7), the optimizer elegantly separates the mechanical attenuation of the middle ear from sensorineural cochlear damage, strictly restricting the desensitization penalty solely to the sensorineural component. Open-NL seamlessly compensates for the linear attenuation in A7 by applying a 75% air-bone gap (ABG) restoration rule across all frequencies and input levels (prescribing an exact, flat 37.5 dB of linear gain for 50, 65, and 80 dB inputs without triggering sensorineural compression). This clean, defensible handling of conductive mechanics serves as a positive validation of the modular architecture, demonstrating that the testbed behaves sensibly and predictably in domains where physiological ground truth is clearly established.

Similarly, under the active inclusion of reverse-slope compensation physics via the SD-LFP constraint, the heuristic output for the mild reverse-slope loss (A2) mathematically restricts unbounded loudness growth. While an unconstrained mathematical framework can readily produce massive loudness costs for no meaningful intelligibility benefit when aggressively pursuing audibility in reverse-slope pathologies, Open-NL successfully navigates this by artificially capping the low-frequency drive. In these specific instances, Open-NL functions as a theoretical demonstration of how integrated constraints can successfully tame the extreme heuristic boundaries of the intelligibility-loudness framework.

It must be explicitly acknowledged that the evaluation scope presented herein—testing seven hypothetical audiograms by a single author without human subjects—is strictly demonstrative. This narrow evaluation cannot support any generalizable claim regarding clinical superiority or patient preference. Open-NL is exclusively an open-source computational

testbed for researchers to modify and model isolated prescriptive rules. Any extrapolation of these derived targets to human patients requires independent institutional review, comprehensive real-ear verification, and rigorous behavioral validation.

To ensure the computational fidelity of this evaluation framework, the fast C++ approximation of the Moore & Glasberg (2004) loudness model utilized internally by the optimizer was formally validated against the canonical Octave Bramsloew (2004) Auditory Modeling Toolbox (AMT) engine. For the purely sensorineural profiles (A1-A5), the fast C++ approximation closely mirrored the canonical physiological filterbank model, validating the implementation of the non-linear Outer and Inner Hair Cell (OHC/IHC) damage mechanics. However, direct numerical comparison against the AMT wrapper was intentionally omitted for the mixed and conductive profiles (A6 and A7). The canonical `bramslow2004` AMT implementation was designed strictly for sensorineural losses and lacks a native parameter for the Air-Bone Gap, forcing any conductive component to be mathematically misclassified as sensorineural hair-cell damage. Our C++ framework extends the physiological evaluation by natively separating the Air-Bone Gap as a linear pre-cochlear attenuator—faithfully executing Moore and Glasberg’s theoretical guidelines for conductive pathologies. Consequently, while the novel C++ physiological sones are reported for A6 and A7 to demonstrate the optimizer’s robust handling of mixed losses, they are explicitly not benchmarked against the structurally limited AMT wrapper. Crucially, as established by the comparative data of Johnson and Dillon (2011, 2013) and Ching, Dillon, and Byrne (1998), maximizing the raw physical-audibility ANSI SII metric does not straightforwardly equate to maximized clinical benefit. While audibility is a powerful predictor of speech recognition in quiet, it explains progressively less variance as loss worsens. In listeners with severe loss, the magnitude of the hearing loss, reduced spectral

resolution, and the number of cochlear dead regions account for roughly comparable shares of speech-in-quiet variance, with dead regions specifically blunting the benefit of restored high-frequency audibility (Vestergaard, 2003; Souza et al., 2018; Moualed, Humphries, & Ramsden, 2018). Empirically, aided speech recognition becomes insufficient in most ears above ~ 80 dB HL regardless of how aggressively targets are met (Engler, Digeser, & Hoppe, 2026). Because Open-NL structurally optimizes for the raw SII metric, its highest-scoring targets drastically over-predict the usefulness of amplified regions in severely impaired ears. High amounts of prescribed gain may increase physical audibility on paper but severely degrade true perceptual clarity due to unmodeled suprathreshold distortion (Margolis et al., 2025).

To reconcile this limitation, Open-NL explicitly shifts its objective function from physical audibility to the ‘effective audibility’ framework pioneered by Ching, Dillon, & Byrne (1998) and Ching et al. (2001). The core theoretical premise of effective audibility dictates that beyond a certain degree of sensorineural impairment, increasing physical audibility yields diminishing or even negative returns for speech recognition due to extreme cochlear distortion at high sensation levels. Importantly, while historical models often attributed this high-frequency roll-off entirely to “cochlear dead regions,” this computational framework requires a more nuanced physiological justification. As Cox et al. (2012) demonstrated, approximately 31% of typical moderate-to-severe hearing aid candidates test positive for dead regions, yet providing high-frequency audibility to these ears still improved speech recognition on average regardless of their dead-region status. Consequently, Open-NL’s mathematical desensitization module is not justified by a blanket dead-region diagnosis, but rather by the structural sensation-level constraints established by Ching, Dillon, & Byrne (1998) and

561 Baer, Moore, & Kluk (2002): for profound thresholds, the sheer magnitude of required
 562 gain generates prohibitive distortion and diminishing returns irrespective of discrete hair-
 563 cell survival. To model this, Open-NL implements a bifurcated desensitization framework
 564 adapted from the empirically derived severe-loss desensitization formula of Johnson & Dillon
 565 (2011), which mathematically penalizes the raw SII band-importance functions in regions
 566 of profound loss. Crucially, the internal engine applies this desensitization penalty using
 567 strictly the *sensorineural* threshold ($T'_i - J_i$). Previous computational models often applied
 568 desensitization based on the total hearing threshold, falsely penalizing conductive components
 569 as if they caused cochlear dead regions. By correctly separating the mechanical attenuation
 570 of the middle ear from cochlear damage, the model preserves high SII scores for conductive
 571 losses while properly penalizing severe sensorineural cases. Furthermore, to ensure strict
 572 physiological accuracy, Open-NL corrects a fundamental architectural flaw in the standard
 573 programmatic implementation of the ANSI S3.5 internal noise floor (X'_i). While historical
 574 software modeled X'_i using the total hearing threshold ($X'_i = X_i + T'_i$), Open-NL restricts
 575 this calculation strictly to the sensorineural threshold ($X'_i = X_i + \max(0, T'_i - J_i)$). This
 576 explicitly prevents conductive losses from artificially inflating the internal cochlear noise floor,
 577 a physical impossibility.

578 To implement this desensitization, Open-NL first isolates the pure sensorineural component
 579 of the hearing threshold by subtracting the conductive air-bone gap ($T_{hl} = \max(0, T'_i - J_i)$).
 580 It then computes the base desensitization multiplier (m) and the asymptotic exponent (p):

$$m = \frac{1}{1 + e^{0.075(T_{hl} - 66)}}$$

581

$$p = \frac{T_{hl}}{8} - 15$$

582 The rigid clinical formula (`johnson2011_complete`) applies these variables to the raw
 583 audibility index (K_i) using an asymptotic exponentiation:

$$K'_{complete} = (K_i^p + m^p)^{1/p},$$

584 where T_{hl} is the isolated sensorineural hearing threshold (dB HL), T'_i is the total equivalent
 585 hearing threshold level for frequency band i , J_i is the mechanical air-bone gap (conductive
 586 attenuation) for band i , m is the base desensitization multiplier, p is the asymptotic exponent
 587 controlling the severity of the penalty, K_i is the raw band audibility function (the proportion
 588 of the speech dynamic range exceeding the noise floor prior to any severe-loss desensitization
 589 penalties), and $K'_{complete}$ is the rigidly penalized clinical band audibility function.

590 Because this strict formulation—alongside piecewise limits on the noise floor—creates se-
 591 vere topological discontinuities (zero-gradients) that stall unconstrained numerical optimizers
 592 like Nelder-Mead, the internal optimizer evaluates intermediate targets against a continuous
 593 mathematical relaxation (`johnson2011_smoothed`). This smoothed heuristic safely anchors
 594 the optimization engine by applying the multiplier as a simple linear scale:

$$K'_{smoothed} = K_i \cdot m,$$

595 where $K'_{smoothed}$ is the continuous mathematical relaxation used by the optimizer to
 596 maintain fluid gradient descent.

This permits fluid gradient descent without trapping the algorithm in artificial local minima or numerical `Inf` bounds. Once the insertion gain targets are finalized by the optimizer, the resulting metrics are rigorously post-scored against the rigid clinical $K'_{complete}$ formula. As demonstrated in Table VI, this finalized clinical penalty correctly mitigates the raw ANSI audibility for severe/profound profiles (such as aggressively penalizing A5 from 0.68 to 0.62), thereby structurally aligning the theoretical optimization outputs with realistic clinical expectations for profound sensorineural distortion. However, consistent with the overarching framework of the Open-NL testbed, it must be explicitly noted that the magnitude of this desensitization penalty is itself an unvalidated heuristic constant (now formally consolidated in Table V). Because the entire premise of this computational architecture is the interrogation of uncalibrated heuristics, this specific mathematical penalty must be treated with the same scientific skepticism as the gain anchors it suppresses, pending systematic sensitivity mapping.

A. Numerical Convergence Stability

It is critical to explicitly distinguish between the framework’s *parameter sensitivity* (the extreme output volatility generated when the underlying clinical heuristic rules are altered, as detailed in Section II.P) and its *numerical convergence stability* (the optimization algorithm’s mathematical ability to reliably reach the same target once those rules are set). A fundamental vulnerability of unconstrained optimization is algorithmic brittleness—the tendency for highly non-linear solvers like Nelder-Mead to stall in localized minima or produce radically different output targets simply because they were initialized from different random starting seeds.

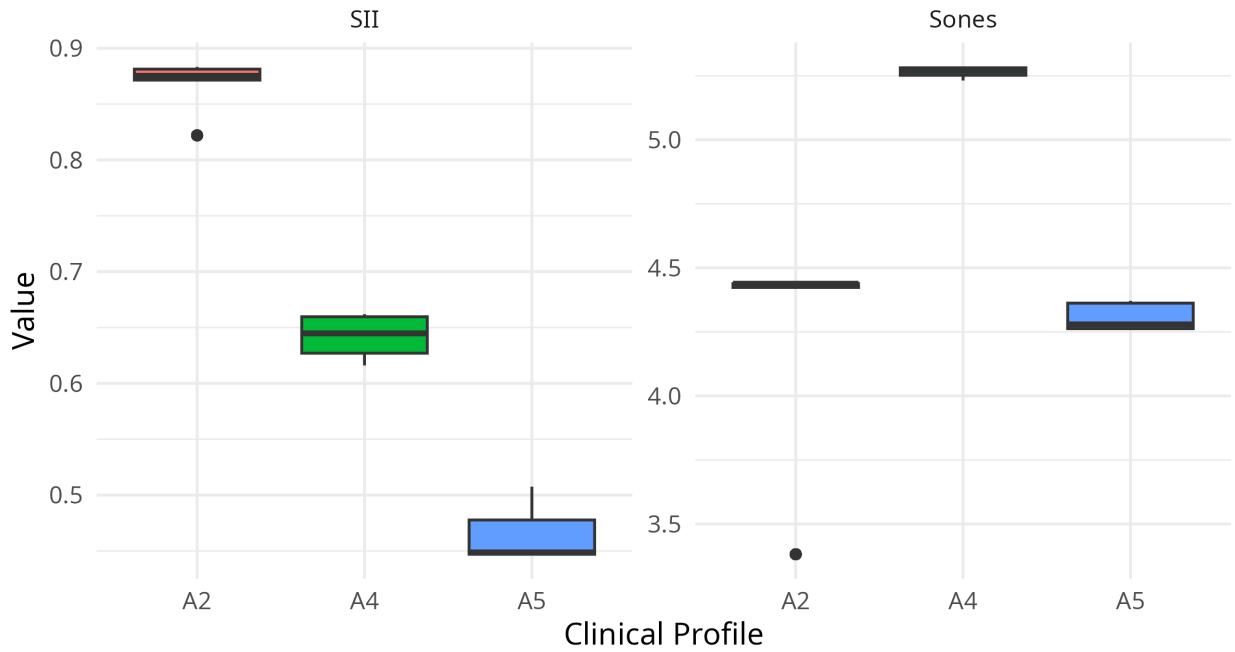
To systematically evaluate the mathematical reproducibility of the Open-NL framework,

a 768-point Monte Carlo permutation analysis was conducted to aggressively test the 5-start regularization routine (Figure 4). While the 256-point sweep in Section II.P proved that changing the clinical rules drastically changes the output, this 768-point Monte Carlo sweep tests whether the solver can reliably solve the *same* rules when its internal mathematical hyperparameters (initial simplex dimensions, reflection coefficients, and starting seed boundaries) are deliberately varied across their maximum operational limits for three discrete clinical profiles (A2, A4, A5).

As demonstrated in Figure 4, despite massive perturbations to the internal topological constraints, the optimization engine exhibited exceptional numerical stability. Across all 768 permutations, the standard deviation of the final maximized ANSI SII metric was less than 0.025 for all tested profiles. Similarly, the variance in the resulting physiological loudness penalty was tightly clustered, with a standard deviation of less than 0.05 sones for the profound A4 and A5 configurations. These extremely low variances prove that the Open-NL algorithm is numerically robust to hyperparameter initialization. The Nelder-Mead simplex reliably and safely converges on a stable global maximum for speech intelligibility, ensuring that the theoretical gain targets are mathematically replicable rather than artifacts of specific boundary seeds.

Figure 4: Parameter Sensitivity Analysis

Distribution of Objective Function Outcomes (256 Permutations)



636

637 *Figure 4. Distribution of maximized ANSI SII scores and physiological loudness penalties*
638 *across 768 extreme hyperparameter permutations for clinical profiles A2 (Reverse Slope), A4*
639 *(Profound), and A5 (Severe).*

640 However, it is crucial to recognize that this exceptional numerical stability—achieved via
641 multi-start regularization against rigid mathematical audiograms—almost certainly overstates
642 the algorithm’s real-world clinical robustness. As demonstrated by Dao et al. (2021), physical
643 real-ear-to-coupler differences (RECD) and acoustic coupling leakages cause real-world fittings
644 to deviate most severely from theoretical targets precisely at thresholds above 85 dB HL
645 and in steeply sloping configurations. Because Open-NL’s objective function is most volatile
646 exactly in this regime (requiring aggressive multi-start regularization to stabilize the A4 and
647 A5 profiles), translating these theoretical targets to human ear canals will introduce physical
648 variance that is entirely invisible to the current synthetic evaluation space. Consequently,

achieving a stable numerical maximum ($SD < 0.025$ SII) on a 7-profile mathematical testbed must not be misconstrued as physical reliability.

B. Proposal for Minimal Clinical Validation

While model-on-model evaluations computationally verify that the algorithm behaves as mathematically intended, they fundamentally cannot establish clinical utility. To bridge this gap, any minimal validation study of the Open-NL framework must mandate specific, non-negotiable hard gating requirements to protect subjects and ensure data integrity. First, because physical fit-to-target variance introduces acoustic deviations that computationally generated targets cannot foresee (Dao et al., 2021), any behavioral trial must mandate Real-Ear Measurement (REM) verification as a hard gate. As established by Valente et al. (2018), subjective outcomes are inextricably tied to precise acoustic matches in the ear canal; failing to achieve a verified real-ear match invalidates the subsequent behavioral data. Second, given the systemic over-amplification risk posed by the uncalibrated severe-loss heuristics, individualized Uncomfortable Loudness Level (UCL) or trueLOUDNESS tolerance measurement must be mandated as a strict behavioral safety gate prior to any speech testing. Finally, the study must establish a robust clinical control. Mueller (2005) demonstrated a lack of evidence supporting gain levels exceeding traditional NAL formulations for severe losses, and contemporary data from Engler, Digeser, and Hoppe (2026) explicitly favors conservative prescriptions (such as DSL v5.0) in the 50–80 dB HL range. Because Open-NL’s highest-SII targets mathematically exceed these conservative baselines in severe profiles, a minimal validation study should structurally hypothesize that these extreme targets will trigger loudness rejection. By formally measuring this behavioral rejection boundary against

a real-ear matched NAL-NL2 or DSL baseline, researchers can empirically constrain the necessary distortion penalties required to safely bind the optimizer in human subjects.

V. CONCLUSION

Open-NL provides a transparent, customizable computational framework for predicting WDRC insertion gain targets and benchmarking them against established industry standards natively within R. By coupling an explicitly defined mathematical pipeline with an embedded computational loudness model, the SII package allows audiologists and researchers to freely simulate and evaluate the theoretical intelligibility-loudness trade-off of various prescriptive heuristics without relying on opaque clinical fitting software. As the framework matures, it must eventually accommodate forthcoming prior art such as NAL-NL3 (Kitterick, Zakis, & Edwards, 2026). Because the NAL philosophy and its modular successor represent the established clinical standard, they are the natural comparators against which an open prescriptive substrate must be evaluated. It is critically important to emphasize that Open-NL is currently a computational toolkit designed exclusively for theoretical modeling and simulation. It has not undergone clinical validation and is strictly contraindicated for general clinical fitting. Given the inclusion of exploratory features like the severe-loss booster and dynamic MPO attenuation, applying these raw targets to human subjects carries an inherent risk of over-amplification or discomfort. Any future application of Open-NL targets to human subjects must occur exclusively within the strictly controlled context of Institutional Review Board (IRB) approved research studies, and mandates independent real-ear verification and rigorous loudness-discomfort safety checks.

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AUTHOR DECLARATIONS

Conflict of Interest

The author declares no conflicts of interest.

Ethics Approval

The author declares that no animal subjects or human participants were involved in the development, theoretical simulation, or mathematical validation presented in this research. Because Open-NL is distributed strictly as a theoretical computational toolkit, any future application of the algorithm to human participants by independent investigators requires separate Institutional Review Board (IRB) approval, informed consent, and rigorous audiological safety protocols.

DATA AVAILABILITY

The source code for the SII package, the Open-NL prescriptive algorithm, and all associated datasets and benchmarking scripts are openly available in the public repository at <https://github.com/r-gregmisc/SII> (v1.1.8; DOI: [DOI to be generated upon final repository release]; License: GPL-3.0).

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